CM3070 Project
Lecture 4

BSc and Diploma in Computer Science and Related Subjects

Synopsis for Lecture

- Reporting Project Work & Progress
- Developing the Project Prototype Report
- Developing the Testing Report
- Developing the Final Project Report
- Advice from Past Students
- Final Things for this Session

Reporting Project Work & Progress

Assessment of the Project module (with formative feedback and peer-assessment)

Description	Coursera Week	Elements	Weightage
Project Proposal (Video)	4	• Project Concept and Related Work *(by 10 Nov 2025)*	Accepted / Not Accepted
Draft Literature Review (Report)	6	• 4-6 examples of previous work or academic literature related to the project	Formative; Peer-assessment
Project Design (Report)	8	• Overview of the project • Name which template used for the project; Domain and users of the project; Design choices, technologies and methods • Overall structure of the project; Project plan of your work including testing and evaluation	Formative; Peer-assessment
Preliminary Project Report	11	• Introduction: this will explain the project concept and motivation for the project • Literatures review: a revised version from peer review • Project Design: revised version from peer review • Feature Prototype: implementation of at least one of the most important technical features used in the project *(by 18 Dec 2025)*	10%
Project Prototype	12	• Written description of the prototype you are developing	Formative; Peer-assessment
Project Testing	14	• 3 questions for the reviewers to answer about your project	Formative; Peer-assessment
Draft Project Report	16	• Introduction: this will explain the project concept and motivation for the project • Literatures review: a revised version from peer review • Project Design: revised version from peer review • Implementation: major algorithms/techniques used, explanation of most important parts of code and a visual representation of results • Evaluation: unit testing, user studies or testing on data and give the results • Conclusion: Short summary of the project and suggest further work	Formative
Project Testing 2	18	• Revised submission of Project Testing	Formative; Peer-assessment
Draft Project Report 2	18	• Revised submission of Draft Project Report	Formative
Unseen Project Examination (2-hour written)	22/23	• Understanding of Project and How it was carried out *(by 16 Mar 2026)*	20%
Final Project Report and Code + Project Presentation Video	24	• Project presentation video and the final project report and code *(by 8 Mar 2026)*	65%
Check in Quizzes / Progress Logs	1-20	• Online Quizzes	5%
Total Weightage			**100%**

Project Prototype (Coursera Week 12 Submission)

Developing the Project Prototype (1/3)

- Choose an appropriate approach for your own project. Justify why you have chosen the development approach. Think about how you will evaluate your chosen development approach at the end of the project.
- By Coursera Week 12, students are required to submit a prototype for a peer review.
- In this assignment, students will submit a written description of the prototype. It should include:
  - The idea for the project
  - Name which template you have used for the project (markers need to cross-refer to the templates)

Developing the Project Prototype (2/3)

- In this assignment, students will submit a written description of the prototype. It should include: (cont'd)
  - An overview of the project and an explanation of how this prototype fits in to the project as a whole.
  - A description of the features being implemented in the prototype.
  - An explanation of the algorithms, techniques and methods used. Students are encouraged to use diagrams to explain them.
  - An explanation of the code. Students should not include entire code but should quote important parts and explain them (do not just paste in code without explaining it).

Developing the Project Prototype (3/3)

- In this assignment, students will submit a written description of the prototype. It should include: (cont'd)
  - A visual representation of the product or results. This could be a screenshot of the user interface or graphs and tables for a data science project.
  - An evaluation of how successful the prototype is and how you intend to improve it.
- The write up should take:
  - The form of a PDF document of maximum 4 pages.
  - Minimum font 11pt, and minimum margins 2cm.
  - Students can have additional pages of images or references that do not contribute to the page count.

Marking Criteria for Project Prototype

- The Feature Prototype will be marked according to the following criteria:
  - Is the report clearly written and presented?
  - Are the diagrams and images appropriate and clear? Did you explain them?
  - Is the project prototype of high quality?
  - Is the project prototype technically challenging?
  - Does the report successfully evaluate the project prototype show suitable improvements if appropriate?
  - Is the code clear and readable? Did you explain them?
  - Is the code of high quality? Coding techiques?

Testing Peer Review (Coursera Week 14 & Week 18 Submission)

Verification and Validation your application / model

- The aim is to detect and remove errors in the application or model.

  - ✓ Are we building the Product right?
  - ✓ Are we building the right Product?

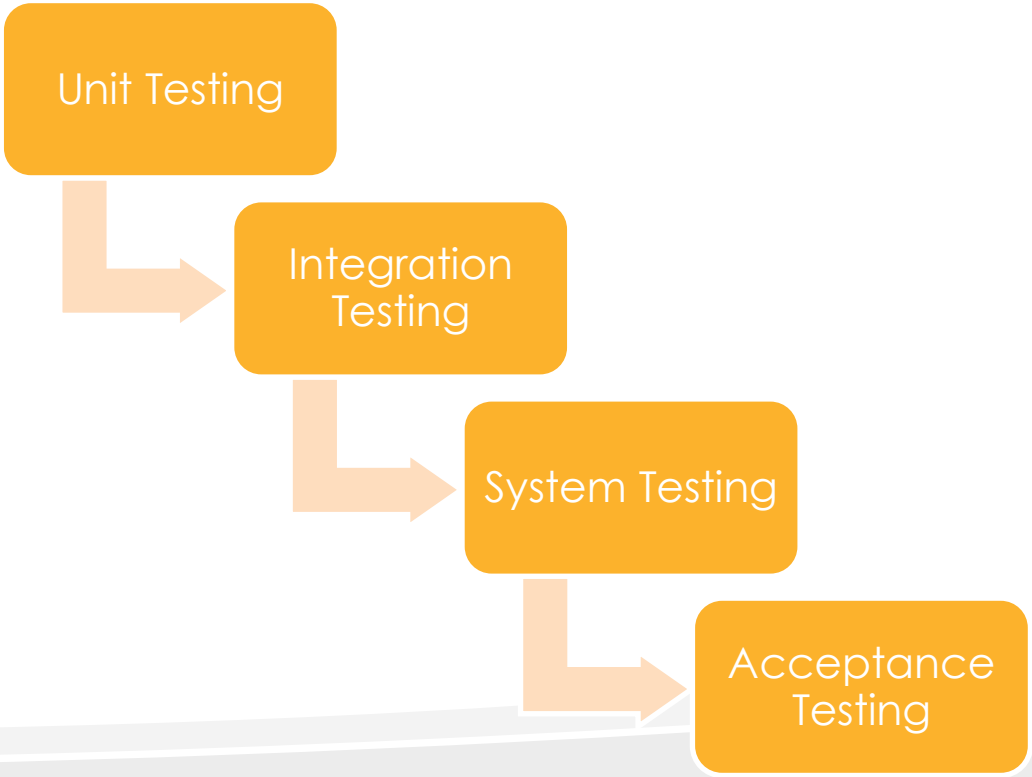
- Verification and Validation are concerned with addressing this issue.
- Students should hope to do this as early as possible.

Verification vs Validation

Considerations	Verification	Validation
	Develop the product RIGHT	Develop the RIGHT product
Focus	Ensuring that the product has been developed correctly	Ensuring that the product meets user requirements
Evaluation	Adherence to design specifications, standards, and guidelines	Fulfillment of user needs and expectations
Activities	Code reviews, inspections, static analysis, walkthroughs, desk-checking, white box testing	Functional testing (black box testing), user acceptance testing, performance testing
Timing	Throughout the development lifecycle	Typically towards the end of the development process
Defect Detection	Early detection of defects to prevent propagation	Identification of gaps between user expectations and the final product
Purpose	Catching and fixing defects in the software development process	Confirming the software's fitness for its intended purpose
Outcome	Improved adherence to standards and specifications	Validation of software quality and value to stakeholders

Applicable Testing for your Project

- In any development work, you should minimally focus on the following types of testing:



Unit Testing

1. Unit Testing

It focuses on the smallest unit of software design. In this, we test an individual unit or group of interrelated units. It is often done by the programmer by using sample input and observing its corresponding outputs.

Example:

```
a) In a program we are checking if loop, method or
   function is working fine
b) Misunderstood or incorrect, arithmetic precedence.
c) Incorrect initialization
```

Reference: https://www.geeksforgeeks.org/types-software-testing/?ref=leftbar-rightbar

Integration Testing

2. Integration Testing

The objective is to take unit tested components and build a program structure that has been dictated by design. Integration testing is testing in which a group of components is combined to produce output.

Integration testing is of four types: (i) Top-down (ii) Bottom-up (iii) Sandwich (iv) Big-Bang

Example

```
(a) Black Box testing:- It is used for validation.
In this we ignore internal working mechanism and
focus on what is the output?.

(b) White Box testing:- It is used for verification.
In this we focus on internal mechanism i.e.
how the output is achieved?
```

Reference: https://www.geeksforgeeks.org/types-software-testing/?ref=leftbar-rightbar

System Testing

7. System Testing

This software is tested such that it works fine for the different operating systems. It is covered under the black box testing technique. In this, we just focus on the required input and output without focusing on internal working.

In this, we have security testing, recovery testing, stress testing, and performance testing

Example:

```
This include functional as well as non functional
testing
```

Reference: https://www.geeksforgeeks.org/types-software-testing/?ref=leftbar-rightbar

Acceptance Testing

It is a formal testing according to user needs, requirements and business processes conducted to determine whether a system satisfies the acceptance criteria or not and to enable the users, customers or other authorized entities to determine whether to accept the system or not.

Acceptance Testing is the last phase of software testing performed after System Testing and before making the system available for actual use.

Reference: https://www.geeksforgeeks.org/acceptance-testing-software-testing/?ref=lbp

Developing the Testing Peer Review (1/3)

- In week 14 students are required to submit a testing for a peer review. This peer review is each student's opportunity to get feedback from other students about the project according to your own evaluation criteria.
- Students will submit three questions that you would like the reviewers to answer about the project. The reviewers will answer the question on a scale: agree, partially agree, neither agree nor disagree, partially disagree, disagree.
- Each student's questions should be suitable for an answer of this form. For example, "I could easily understand how to use the interface", "The search results were appropriate for my query".

Developing the Testing Peer Review (2/3)

- Each student should submit your project in a form that other students can use, for example, a built application.
- You can assume that students will have access to software used in our modules (e.g. python, or unity), but that might limit reviewers to students who have done those modules.
- E.g. if you are developing a game project you might want to submit a webGL build as it is accessible to more students than a unity project would be).

Developing the Testing Peer Review (3/3)

- If it is not possible to submit a runnable project (e.g. if you are developing hardware), you can submit a video.
- Students should also provide written instructions that clearly explain what software and hardware are required to run the project.

Potential Questions (1/3)

1. What are the testing objectives? Understanding the testing objectives helps define the purpose and scope of the testing effort. Questions to ask include:
   - What specific aspects of the software or system need to be tested?
   - What are the goals and expectations for the testing process?
   - Are there any specific quality attributes or metrics that need to be evaluated?

Potential Questions (2/3)

2. What testing techniques and approaches should be employed? Selecting appropriate testing techniques is essential to ensure comprehensive and effective test coverage. Consider the following:
   - Which types of testing, such as functional, performance, security, or usability, should be conducted?
   - Are there any specific testing methodologies, such as black-box, white-box, or gray-box testing, that should be used?
   - Should automated testing tools or manual testing techniques be employed?

Potential Questions (3/3)

3. How will testing be measured and evaluated? Defining criteria for measuring the effectiveness and success of testing is important. Consider the following aspects:
   - What are the desired quality metrics, such as defect detection rate, test coverage, or reliability?
   - How will the test results be documented, analysed, and reported?
   - Are there any specific exit criteria or thresholds for test completion and sign-off?

Draft Final Report (Coursera Week 16 & Week 20 Submission)

Developing the Final Report (1/5)

- By Coursera Week 16 and 20, students are required to submit a Draft report and Draft report 2.
- In this review, the students will submit a draft of some aspects of the Final Project Report.
- The report will consist of 6 mandatory chapters (excluding Title Page, Appendices and References). Besides the page limits listed, you can have additional pages of images and references.
- You can return to the previous peer reviews for instructions on the first four parts of this submission.

Developing the Final Report (2/5)

Title page	Giving the title, student name, date, degree title, supervisor name, name of local institution attended (or write 'Self-study'), University of London International Programmes student registration number
Introduction*	This will explain the project concept and motivation for the project (this chapter can be elaborated from your project proposal previously submitted). This must also state the idea for the project and which project template you are using (1000 words)

Developing the Final Report (3/5)

Literature Review*	This is a revised version of the document that students previously had submitted for the second peer review (2500 words).
Project Design*	This is a revised version of the document that students previously had submitted for the third peer review (2000 words).

Developing the Final Report (4/5)

Implementation*	This should describe the implementation of the project. This should follow the style of the Coursera Week 12 and 14 peer reviews (Project Prototype and Test Plan but must expand to cover your full implementation), describing the major algorithms/techniques used, explanation of the most important parts of the code and a visual representation of the results (e.g. screenshots or graphs). (2000 words)
Evaluation*	Describe the initial evaluations that you have carried out (e.g. unit testing, user studies or testing on data) and give the results. This should give a critical evaluation of the project so far making clear what you have achieved and what you can improve. (1500 words before Mar 2025, 2500 words after Mar 2025)

Developing the Final Report (5/5)

Conclusion*	This can be a short summary of the project as a whole but it can also bring out any broader themes you would like to discuss or suggest further work. (1000 words)
Appendices	Appendices should be cross-referenced in the relevant section of the main text. No Appendix should be present unless cross-referenced from the main text.
References	Giving full publication details of all literature referred to in the FYR. For details, please refer to: https://onlinelibrary.london.ac.uk/support/information-skills/organising-and-citing-your-references/citing-references. This list will generally be much shorter than that for the (later and much larger) Final Project Report.

Word Limit for Final Report

- A big advantage of word limits is that using this approach makes the practice of including your figures and diagrams in appropriate and relevant places in the report much easier.
- Word limits (Total word max: 9500)
  - Introduction: 1000 words
  - Literature review: 2500 words
  - Design: 2000 words
  - Implementation: 2000 words
  - Evaluation: 2500 words
  - Conclusion: 1000 words
- NOTE that if you add up all of the above, this comes to more than 9500 words. 9500 words is a strict limit, and each of the chapter's limit is strict, but this approach allows you to exercise some flexibility in the spread of your writing across the different chapters, to suit your particular project).

Plagiarism – A Warning

How to avoid plagiarism (1/3)

- Plagiarism is serious academic misconduct, involving:
  - submitting a piece of coursework which contains unacknowledged material from books or other sources;
  - copying any part of the work of fellow students;
  - reproducing any part of a piece of coursework completed in previous years without acknowledgment.

- Please note that plagiarism constitutes an examination offence. If the University confirms a piece of coursework to be plagiarised:
  - a mark of zero (0) will be given for the coursework concerned
  - you will be formally reprimanded.

Reference Source (16 Mar 2024): https://computing.elearning.london.ac.uk/mod/page/view.php?id=5176

How to avoid plagiarism (2/3)

- DO use Turnitin to check your work before submitting your assignments.

- Repeated offences of plagiarism will attract greater penalties. The University may decide:
  - that there should be a mark of zero for the course concerned
  - that there should be a mark of zero for all of the courses taken in the year concerned
  - that you be excluded from any or all future examinations of the University.

Reference Source (16 Mar 2024): https://computing.elearning.london.ac.uk/mod/page/view.php?id=5176

How to avoid plagiarism (3/3)

- A reminder to all students:
  - it is not sufficient to provide a list of sources as this does not distinguish your work from the source material. No credit will be given as a plagiarism offence has been committed since the copied material has not been identified.
  - any quoted material (that is, taken verbatim or word for word) must be clearly identified: quotation marks should be used (along with italics and indentation for clarity) and author surname, date of publication and page number(s) cited. You should avoid quoting at length from any source, as this does not demonstrate your understanding.
  - if material is paraphrased (that is, re-worded) the extent of the paraphrase must be clear, and author surname and date of publication cited. It is not acceptable to retain significant amounts of the original wording.
  - full details of all sources identified in the text should be included in a separate alphabetical referencing list at the end of the assignment. See the Harvard Referencing Guide for further details.
  - Graphical material should also be acknowledged.

Reference Source (16 Mar 2024): https://computing.elearning.london.ac.uk/mod/page/view.php?id=5176

Delivering a good Final Project Report

Aspects of a good Project Report

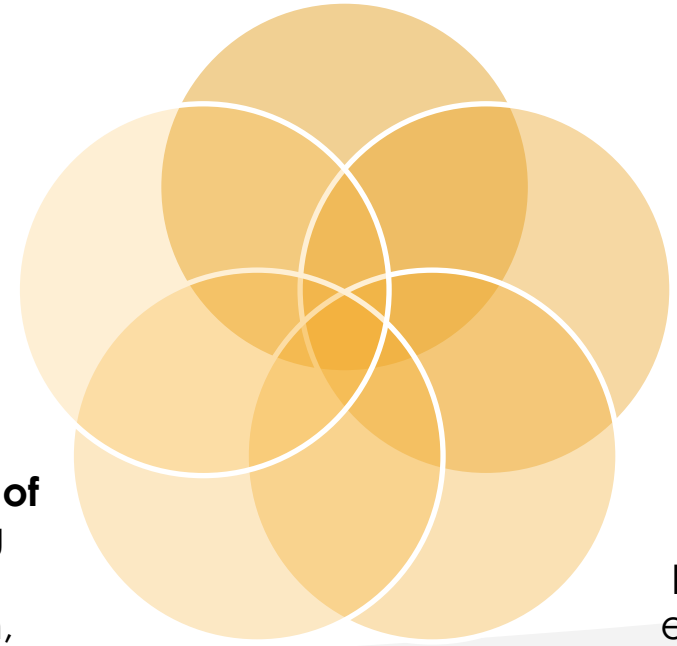
1. Structure – Conformance to standards

2. Academic standards – Proper cross-referencing and Reference list

3. Depth of knowledge – The extent of the work undertaken in the time permitted

4. Completeness of report – Making sure the implementation, testing and evaluation are documented

5. Report quality – Good clear English. Appropriate use of diagrams and tables



Nine Basic Ways to Improve Your Style in Academic Writing

1. Use ACTIVE VOICE
2. Mix it up in terms of PUNCTUATION
3. Vary your SENTENCE STRUCTURE
4. Closely related to this, avoid CHOPPINESS
5. Avoid REPETITION
6. Be CONCISE
7. Use the VOCABULARY that you know
8. But also work on expanding your VOCABULARY
9. Keep language FORMAL and avoid language of everyday speech

Adapted from: https://slc.berkeley.edu/writing-worksheets-and-other-writing-resources/nine-basic-ways-improve-your-style-academic-writing

Advice from past students

Academic Writing

Ensure that the report is at least 99% **mistake free in the sense of grammar, syntax, plagiarism** etc. as it is the first thing to be seen by the markers of the module before anything else, regardless of whether you have a perfect prototype working. Treat the markers as idiots, **do not assume they would know everything you write**. What it may be in Singapore, may not be the same meaning in the UK. For those who have difficulty writing their reports, they should **reference previous years' project reports on how they were written**. Ensure that the markers have an easy time to read your report.

Developing Prototypes

Those **attempting to do a prototype** of their project would be recommended to do something **within their limits of programming**. The complexity of the project does not have to be difficult, but more of an idealistic concept that would be practical or out of the box. In the worst case scenario, if time runs out, I would recommend the students to **at least do the interface out to show the screen flow**, rather than just a report without one, as the markers might not be able to visualize what the student is trying to achieve.

Time Management

As most of my predecessors would say, try **not to leave the project to the last few months**. A rush work is never a quality work. Ensure you **meet deadlines set by UOL**, or personal deadlines, with **buffer timing to safeguard yourself** as the project can make a hell lot of difference between a 2nd Upper Class and a 2nd Lower Class, especially if your grades are not up to mark.

Final Things for this Session

Importance of Test Plans

You need to develop a test plan with an objective and outcomes. It is necessary to document the types of testing applicable to your project and what will be tested.

See you at next session !